

20.2 Leverage and the hat matrix

Section 6.8 proved the basic properties of the leverage h_{ii} : it lies in $[0, 1]$, the leverages sum to p , $h_{ii} \geq 1/n$ with an intercept, and h_{ii} is an affine function of the Mahalanobis distance of $\mathbf{x}_{(i)}$ from the centroid of the regressors (Proposition 6.8.1 and Proposition 6.8.2). This section asks when a leverage deserves attention, how designed and observational data differ, and how to measure the pull of a case on one coefficient.

The fitted value of case i is

$$\hat{y}_i = h_{ii}y_i + \sum_{j \neq i} h_{ij}y_j,$$

so h_{ii} is the weight of a case in its own fitted value. Since $\sum_j h_{ij}^2 = h_{ii}$ (Proposition 6.8.1(a)), a case with h_{ii} near one leaves little weight for the others: its fitted value is essentially its own response, and the fit carries no information about whether y_i is right. Leverage is *potential* influence, a property of the design alone; Section 20.4 combines it with the residual.

20.2.1 How large is large?

The average leverage is p/n . The most widely used rule (Hoaglin and Welsch 1978) calls a case of high leverage if $h_{ii} > 2p/n$; a stricter variant uses $3p/n$. To see what these rules mean, compute how often they fire under a reference design with multivariate normal regressors.

Proposition 20.2.1. Leverage under a Gaussian design

Let $\mathbf{X} = [\mathbf{1}, \mathbf{Z}]$, where the rows of the $n \times k$ matrix $\mathbf{Z}$ are independent $N_k(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ vectors with $\boldsymbol{\Sigma}$ positive definite, and let $n \geq k + 2$. Then $\mathbf{X}$ has full column rank $p = k + 1$ with probability one, and for every i ,

$$\frac{h_{ii} - 1/n}{1 - 1/n} \sim \text{Beta}\left(\frac{k}{2}, \frac{n - 1 - k}{2}\right).$$

Proof. Reduction. Let $\mathbf{C} = \mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top$. By Lemma 6.6.1 with $\mathbf{X}_2 = \mathbf{1}$, $\mathbf{M} = n^{-1}\mathbf{1}\mathbf{1}^\top + \mathbf{P}$, where $\mathbf{P}$ is the orthogonal projection onto $\mathcal{C}(\mathbf{CZ})$, a subspace of $\mathbf{1}^\perp$. So $h_{ii} - 1/n = \mathbf{e}_i^\top \mathbf{P} \mathbf{e}_i$. Replacing $\mathbf{Z}$ by $(\mathbf{Z} - \mathbf{1}\boldsymbol{\mu}^\top)\boldsymbol{\Sigma}^{-1/2}$ does not change $\mathcal{C}(\mathbf{CZ})$, because $\mathbf{C}\mathbf{1} = \mathbf{0}$ and $\boldsymbol{\Sigma}^{-1/2}$ is nonsingular. So we may assume that the entries of $\mathbf{Z}$ are independent $N(0, 1)$. Then $\mathbf{CZ}$ has rank k with probability one (Exercise (C1)), and so does $\mathbf{X}$.

Invariance. Let $\mathbf{v} = \mathbf{C}\mathbf{e}_i / \|\mathbf{C}\mathbf{e}_i\|$, a unit vector in $\mathbf{1}^\perp$, with $\|\mathbf{C}\mathbf{e}_i\|^2 = 1 - 1/n$. Since $\mathbf{P} = \mathbf{P}\mathbf{C} = \mathbf{C}\mathbf{P}$,

$$h_{ii} - \frac{1}{n} = (\mathbf{C}\mathbf{e}_i)^\top \mathbf{P} (\mathbf{C}\mathbf{e}_i) = \left(1 - \frac{1}{n}\right) \mathbf{v}^\top \mathbf{P} \mathbf{v}.$$

Write $\mathbf{P}(\mathbf{Z})$ to show the dependence on $\mathbf{Z}$. Let $\mathbf{w}$ be any other unit vector in $\mathbf{1}^\perp$, and let $\mathbf{H} = \mathbf{I} - 2\mathbf{a}\mathbf{a}^\top / \mathbf{a}^\top \mathbf{a}$ with $\mathbf{a} = \mathbf{v} - \mathbf{w}$ (or $\mathbf{H} = \mathbf{I}$ if $\mathbf{w} = \mathbf{v}$). This is a symmetric orthogonal matrix, and $\mathbf{H}\mathbf{v} = \mathbf{w}$ because $\|\mathbf{v}\| = \|\mathbf{w}\|$. It fixes $\mathbf{1}$, because $\mathbf{1} \perp \mathbf{a}$, so $\mathbf{H}$ commutes with $\mathbf{C}$ and maps $\mathcal{C}(\mathbf{CZ})$ onto $\mathcal{C}(\mathbf{CHZ})$. Hence $\mathbf{P}(\mathbf{HZ}) = \mathbf{H}\mathbf{P}(\mathbf{Z})\mathbf{H}$, and $\mathbf{w}^\top \mathbf{P}(\mathbf{Z})\mathbf{w} = \mathbf{v}^\top \mathbf{H}\mathbf{P}(\mathbf{Z})\mathbf{H}\mathbf{v} = \mathbf{v}^\top \mathbf{P}(\mathbf{HZ})\mathbf{v}$. Each column of $\mathbf{HZ}$ is $N_n(\mathbf{0}, \mathbf{H}\mathbf{H}^\top) = N_n(\mathbf{0}, \mathbf{I})$, and the columns are independent, so $\mathbf{HZ}$ has the law of $\mathbf{Z}$. Therefore the law of $\mathbf{v}^\top \mathbf{P}(\mathbf{Z})\mathbf{v}$ is the same for every unit vector $\mathbf{v} \in \mathbf{1}^\perp$.

Averaging. Let $\mathbf{g} \sim N_n(\mathbf{0}, \mathbf{C})$ be independent of $\mathbf{Z}$, and $\mathbf{U} = \mathbf{g} / \|\mathbf{g}\|$, a random unit vector in

$\mathbf{1}^\perp$. By the previous paragraph, applied conditionally on $\mathbf{U}$, the variable $\mathbf{U}^\top \mathbf{P}(\mathbf{Z})\mathbf{U}$ has the same law as $\mathbf{v}^\top \mathbf{P}(\mathbf{Z})\mathbf{v}$. Now condition on $\mathbf{Z}$ instead. Then $\mathbf{P}$ is a fixed projection of rank k onto a subspace of $\mathbf{1}^\perp$, and $\mathbf{C} - \mathbf{P}$ is a projection of rank $n - 1 - k$ orthogonal to it, so by Theorem 4.4.4 $\|\mathbf{P}\mathbf{g}\|^2 \sim \chi^2(k)$ and $\|(\mathbf{C} - \mathbf{P})\mathbf{g}\|^2 \sim \chi^2(n - 1 - k)$ are independent. Since $\mathbf{g} = \mathbf{C}\mathbf{g}$,

$$\mathbf{U}^\top \mathbf{P}\mathbf{U} = \frac{\|\mathbf{P}\mathbf{g}\|^2}{\|\mathbf{P}\mathbf{g}\|^2 + \|(\mathbf{C} - \mathbf{P})\mathbf{g}\|^2} \sim \text{Beta}\left(\frac{k}{2}, \frac{n - 1 - k}{2}\right)$$

by Exercise (B4). This conditional law does not depend on $\mathbf{Z}$, so it is also the unconditional law. ■

The mean of the beta law is $k/(n - 1)$, which gives $E(h_{ii}) = 1/n + (1 - 1/n)k/(n - 1) = p/n$, as it must. As $n \rightarrow \infty$ with k fixed, $n(h_{ii} - 1/n)$ converges in law to $\chi^2(k)$ (Exercise (B1)), so

$$\Pr\{h_{ii} > cp/n\} \approx \Pr\{\chi^2(k) > c(k + 1) - 1\}.$$

With $k = 3$, the rule $2p/n$ fires for 0.072 of the cases in large samples (0.067 when $n = 50$), and the rule $3p/n$ for 0.012. With $k = 10$ and $n = 500$, the rule $2p/n$ fires for only 0.020. “Twice the average” flags a different share of well-behaved cases for every p : it is a convention, not a test. Figure 20.2.1(a) checks the law against 4000 simulated designs.

```
import numpy as np
from scipy import stats

def leverages(X):
    Q, _ = np.linalg.qr(X)
    return np.sum(Q ** 2, axis=1)

rng = np.random.default_rng(2020)
n, k = 30, 3                                # intercept plus k Gaussian
    regressors
p = k + 1
reps = 4000
h1 = np.empty(reps)
for r in range(reps):
    Z = rng.normal(size=(n, k)) @ np.array([[1.0, 0.0, 0.0], [0.8, 0.6, 0.0], [0.0,
    2.0, 1.0]])
    X = np.column_stack([np.ones(n), Z + 5.0])    # correlated, shifted regressors
    h1[r] = leverages(X)[0]                      # leverage of case 1
B = (h1 - 1 / n) / (1 - 1 / n)                  # should be Beta(k/2, (n-1-k)/2)
law = stats.beta(k / 2, (n - 1 - k) / 2)
print("mean of h_11:", h1.mean(), " theory p/n =", p / n)
print("Kolmogorov-Smirnov p-value against the Beta law:", stats.kstest(B, law.cdf).
    pvalue)
```

[Run this code online](#)

20.2.2 Designed and observational data

In a designed experiment the analyst chooses the rows of $\mathbf{X}$. The largest leverage is at least the average p/n , with equality iff all are equal (Exercise (B2)), and many designs achieve it. If the columns are orthogonal with entries ± 1 , as in two-level factorials, then $\mathbf{X}^\top \mathbf{X} = n\mathbf{I}$ and $h_{ii} = \|\mathbf{x}_{(i)}\|^2/n = p/n$: a 2^3 factorial run twice ($n = 16$) with intercept, main effects and

one interaction ($p = 5$) gives every case leverage $5/16 = 0.3125$. A balanced one-way layout gives every case $1/n_k = p/n$ (Exercise (B2)). Kiefer and Wolfowitz (1960) showed that, over a region of possible design points, the designs maximizing $\det(\mathbf{X}^\top \mathbf{X})$ are exactly those minimizing the largest leverage $\mathbf{x}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}$ over the region, with minimum p/n (for designs viewed as probability measures). For D-optimal designs, good design and small maximum leverage are the same thing.

In observational data the rows are drawn, not chosen, and with light enough tails no case dominates in large samples.

Proposition 20.2.2. Leverage vanishes in large samples

Let the rows $\mathbf{x}_{(1)}, \mathbf{x}_{(2)}, \dots$ be independent copies of a random vector $\mathbf{x}$ with $E\|\mathbf{x}\|^2 < \infty$ and $E(\mathbf{x}\mathbf{x}^\top)$ positive definite, and let $h_{ii}^{(n)}$ be the leverages of the first n rows. Then $\max_{i \leq n} h_{ii}^{(n)} \rightarrow 0$ in probability.

Proof. Write $\mathbf{S}_n = n^{-1} \mathbf{X}^\top \mathbf{X} = n^{-1} \sum_{i \leq n} \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top$ and let λ_n be its smallest eigenvalue. By the strong law of large numbers $\mathbf{S}_n \rightarrow E(\mathbf{x}\mathbf{x}^\top)$ with probability one, and eigenvalues are continuous functions of a symmetric matrix, so $\lambda_n \rightarrow \lambda > 0$, the smallest eigenvalue of $E(\mathbf{x}\mathbf{x}^\top)$. Hence with probability one $\mathbf{X}^\top \mathbf{X}$ is nonsingular for all large n , and the leverages are defined from then on. By Theorem 1.7.7 applied to $(\mathbf{X}^\top \mathbf{X})^{-1}$, whose largest eigenvalue is $1/(n\lambda_n)$,

$$\max_{i \leq n} h_{ii}^{(n)} = \max_{i \leq n} \mathbf{x}_{(i)}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_{(i)} \leq \frac{\max_{i \leq n} \|\mathbf{x}_{(i)}\|^2}{n\lambda_n}.$$

It remains to show that $\max_{i \leq n} \|\mathbf{x}_{(i)}\|^2/n \rightarrow 0$ in probability. For $\epsilon > 0$, by the union bound and Markov's inequality,

$$\Pr\left\{\max_{i \leq n} \|\mathbf{x}_{(i)}\|^2 > \epsilon n\right\} \leq n \Pr\{\|\mathbf{x}\|^2 > \epsilon n\} \leq \frac{1}{\epsilon} E[\|\mathbf{x}\|^2 1\{\|\mathbf{x}\|^2 > \epsilon n\}] \rightarrow 0,$$

by dominated convergence, since $E\|\mathbf{x}\|^2 < \infty$. ■

The condition matters for inference with nonnormal errors. Huber (1973) showed that every fitted value $\mathbf{x}^\top \hat{\boldsymbol{\beta}}$ is asymptotically normal, for every error law with finite variance, iff the largest leverage tends to zero, and Theorem 19.3.2 of Chapter 19 rests on a condition of this Lindeberg type. A case that keeps a fixed share of the leverage keeps a fixed share of its own error in its fitted value, and no averaging makes that error normal.

Figure 20.2.1(b) shows the median, over 300 replications, of the largest leverage with two independent regressors. For normal regressors it falls from 0.322 at $n = 25$ to 0.0030 at $n = 6400$. For t_3 regressors, heavy-tailed but with finite variance, it falls much more slowly, to 0.066. For Cauchy regressors, which have no variance, it does not fall at all (0.857 at $n = 25$, 0.831 at $n = 6400$): one case keeps most of the pull however many are added.

Example. Leverage in Engel's household budgets

Engel's 1857 survey records the annual income and food expenditure of 235 Belgian working-class households. Income is strongly skewed (skewness 2.78): the median is 884 and the

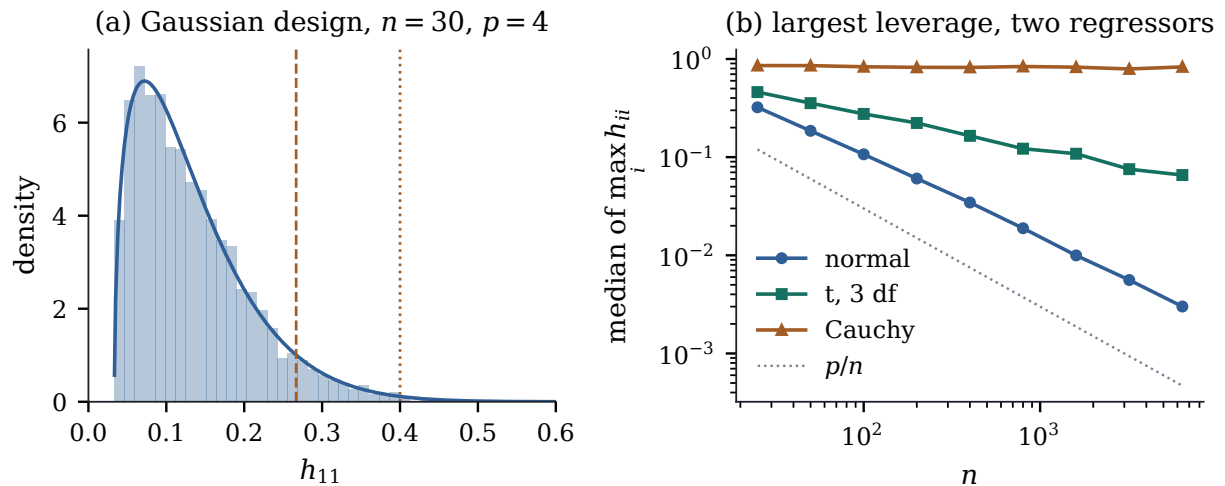


Figure 20.2.1: (a) Leverage of one case in 4000 Gaussian designs ($n = 30$, $p = 4$), the beta density of Proposition 20.2.1, and the cut-offs $2p/n$ (dashed) and $3p/n$ (dotted). (b) Median largest leverage for normal, t_3 and Cauchy regressors.

largest, row 138 of the file, is 4958. In the straight-line regression of food expenditure on income the average leverage is $2/235 = 0.0085$, and the richest household has leverage 0.255, about 30 times the average; the next largest is 0.058. Since $h_{ii} - 1/n = (x_i - \bar{x})^2/S_{xx}$, this one household supplies a quarter (0.251) of the spread of income about its mean, and the fitted line will pass near it whatever it spends on food. On the logarithmic scale its leverage is 0.070, still the largest but no longer in a class of its own. Choosing the scale (Chapter 22) decides which cases the fit depends on.

```
import numpy as np
import statsmodels.api as sm
engel = sm.datasets.engel.load_pandas().data

income = engel["income"].to_numpy()
food = engel["foodexp"].to_numpy()
n = len(income)
for label, x in [("income", income), ("log income", np.log(income))]:
    X = np.column_stack([np.ones(n), x])
    Q, _ = np.linalg.qr(X)
    h = np.sum(Q ** 2, axis=1)
    top = np.argsort(h)[::-1][:3]
    print(f"{label:11s} average p/n = {2 / n:.4f}; largest:",
          ", ".join(f"row {i + 1}: h = {h[i]:.3f}" for i in top))
```

[Run this code online](#)

Leverage also depends on the model: adding a column never decreases a leverage (Exercise (B2)), and a case ordinary for a line can be extreme for a quadratic (Exercise (B3)). A new point inside the range of every regressor can have high leverage because its *combination* of values is unusual (Example (Hidden extrapolation)).

20.2.3 Partial leverage

The leverage h_{ii} measures the pull of case i on the whole fitted surface. The Frisch–Waugh–Lovell theorem isolates one coefficient. Let $\mathbf{M}_{(j)}$ project onto the span of the columns other

than $\mathbf{x}_j$, and let $\tilde{\mathbf{x}}_j = (\mathbf{I} - \mathbf{M}_{(j)})\mathbf{x}_j$.

Definition 20.2.1. Partial leverage

The **partial leverage** of case i for the coefficient of $\mathbf{x}_j$ is

$$\ell_{ij} = \frac{\tilde{x}_{ij}^2}{\|\tilde{\mathbf{x}}_j\|^2}.$$

Proposition 20.2.3. Properties of partial leverage

Let $h_{ii}^{(j)}$ be the leverages of the model without $\mathbf{x}_j$. Then:

- (a) $h_{ii} = h_{ii}^{(j)} + \ell_{ij}$ for every i ;
- (b) $0 \leq \ell_{ij} \leq 1$ and $\sum_i \ell_{ij} = 1$;
- (c) ℓ_{ij} is the leverage of case i in the regression through the origin of $\tilde{\mathbf{y}} = (\mathbf{I} - \mathbf{M}_{(j)})\mathbf{y}$ on $\tilde{\mathbf{x}}_j$, the regression drawn in the added-variable plot for $\mathbf{x}_j$;
- (d) $\hat{\beta}_j = \sum_i (\tilde{x}_{ij}/\|\tilde{\mathbf{x}}_j\|^2)y_i$, so the weight of y_i in $\hat{\beta}_j$ has square $\ell_{ij}/\|\tilde{\mathbf{x}}_j\|^2 = \ell_{ij} \text{Var}(\hat{\beta}_j)/\sigma^2$.

Proof. By Lemma 6.6.1(c) with $\mathbf{X}_1 = \mathbf{x}_j$, $\mathbf{M} = \mathbf{M}_{(j)} + \tilde{\mathbf{x}}_j\tilde{\mathbf{x}}_j^\top/\|\tilde{\mathbf{x}}_j\|^2$. The diagonal entries give (a), and the second term is the hat matrix of the regression in (c). It is a rank-one projection, which gives (b). Part (d) is Theorem 6.6.2(a), together with $\text{Var}(\hat{\beta}_j) = \sigma^2/\|\tilde{\mathbf{x}}_j\|^2$ (Eq. 6.6.2). ■

A case can have high leverage and negligible partial leverage for the coefficient of interest, when its unusual position lies along the other regressors. Partial leverages average $1/n$, so a few times $1/n$ is already large.

Example. Where the District of Columbia's leverage comes from

In the murder-rate regression for the 51 jurisdictions, with poverty, single-parent households and urbanization as regressors, the District of Columbia has leverage 0.506 (Example (High-leverage states)) but partial leverage for poverty only 0.032: its leverage comes from its extreme single-parent and urban percentages, not from its poverty rate. The largest partial leverage for poverty is Alaska's, 0.141. [Section 20.4](#) shows that the District nevertheless moves the poverty coefficient more than any other case, because influence also depends on the residual.

20.2.4 Exercises

20.2.4.1 A. Check your understanding

Exercise. A1

For regression through the origin on one regressor, show that $h_{ii} = x_i^2/\sum_k x_k^2$. Give data

with $n = 10$ in which one leverage exceeds 0.99.

Solution. $\mathbf{M} = \mathbf{xx}^\top / \mathbf{x}^\top \mathbf{x}$, so $h_{ii} = x_i^2 / \mathbf{x}^\top \mathbf{x}$. With $x_1 = 100$ and $x_2 = \cdots = x_{10} = 1$, $h_{11} = 10^4 / (10^4 + 9) > 0.99$.

Exercise. A2

Show that with an intercept the leverages are unchanged when the regressors are replaced by $\mathbf{a} + \mathbf{Bx}$ with $\mathbf{B}$ nonsingular, but not under a nonlinear transformation ([Example \(Leverage in Engel's household budgets\)](#)). Use Proposition 6.8.2.

20.2.4.2 B. Practice

Exercise. B1

In the setting of [Proposition 20.2.1](#), show that $n(h_{ii} - 1/n)$ converges in law to $\chi^2(k)$ as $n \rightarrow \infty$.

Solution. By the proof of [Proposition 20.2.1](#), $h_{ii} - 1/n$ has the law of $(1 - 1/n)U/(U + V)$ with $U \sim \chi^2(k)$ and $V \sim \chi^2(n - 1 - k)$ independent. So $n(h_{ii} - 1/n)$ has the law of $(n - 1)U/(U + V) = U/[(U + V)/(n - 1)]$. By the law of large numbers $V/(n - 1) \rightarrow 1$ in probability, and $U/(n - 1) \rightarrow 0$, so the denominator tends to 1 in probability. Slutsky's theorem gives convergence in law to $U \sim \chi^2(k)$.

Exercise. B2

Show that $\max_i h_{ii} \geq p/n$, with equality iff all leverages equal p/n . Show that if $\mathbf{X}$ has orthogonal columns and every entry is ± 1 , then all leverages equal p/n .

Exercise. B3

Take $x = -3, -2, -1, 0, 1, 2, 3$. Compute the leverage of the case at $x = 0$ for the straight-line model and for the quadratic model $E(Y) = \beta_0 + \beta_1 x + \beta_2 x^2$. Explain the difference.

Solution. For the line, $h = 1/7 + 0^2/28 = 1/7$. For the quadratic, apply [Proposition 20.2.3\(a\)](#) with $\mathbf{x}_j$ the column of squares: $h = 1/7 + \ell$, where ℓ is the partial leverage of the quadratic term. The residual of x^2 on $(1, x)$ is $x^2 - 4$, because x^2 has mean 4 and $\sum_x x(x^2 - 4) = 0$ by symmetry. Its values are 5, 0, -3, -4, -3, 0, 5, with sum of squares 84. So $\ell = 16/84 = 4/21$ and $h = 1/7 + 4/21 = 1/3$. The case at $x = 0$ is the only one near the centre, and in a quadratic model the centre is where the curvature is pinned down.

20.2.4.3 C. Going deeper

Exercise. C1

Complete the proof of [Proposition 20.2.1](#) by showing that if $\mathbf{Z}$ is $n \times k$ with independent $N(0, 1)$ entries and $n \geq k + 2$, then $\mathbf{CZ}$ has rank k with probability one. *Hint:* add columns one at a time, and use that a nonzero normal vector with a nonsingular covariance restricted to a subspace falls in a fixed proper subspace with probability zero.